

Flight Delay Prediction Using Classification Techniques

Name: Dnyaneshwari Maradkar,

Student Id: - 24142497

Dataset: <https://www.kaggle.com/datasets/usdot/flight-delays/data>

Git Hub link- <https://github.com/dnyaneshwari7/flight-delay-prediction-data-mining>

1. Problem Statement

Flight delays continue to be a major issue in the US aviation industry. They affect a very large number of passengers and also create extra costs and operational pressure for airlines and airports. Because of this, being able to identify which flights are more likely to be delayed has clear practical value for both operators and travellers.

This study looks at whether machine learning models can be used to predict flights that are likely to arrive more than 15 minutes late. The prediction is based only on information that is available before departure, such as the airline, origin and destination airports, scheduled departure time, day of the week, and flight distance.

If delays can be predicted in advance, airlines and airports could respond earlier by managing staff and aircraft more effectively, improving communication with passengers, and reducing the way delays spread through the rest of the day's schedule.

2. Data Summary

The dataset used in this study is the US Flight Delays dataset, which is based on flight records from the US Department of Transportation and made available on Kaggle. It contains information on around 5.8 million domestic US flights from 2015. For this analysis, a smaller sample of 50,000 flights was used so that the models could be run more easily and efficiently. After removing cancelled flights and records with missing values, around 46,000 usable flights remained for the final analysis. Key variables used:

- Temporal variables: month, day of the week, and scheduled departure time (this was also converted into departure hour from 0–23 for easier analysis)
- Route variables: origin airport, destination airport, and flight distance
- Carrier variable: airline code (converted into numeric form using label encoding)
- Target variable: delay status — 1 if arrival delay is more than 15 minutes, otherwise

Figure 1 shows the distribution of the two classes in the dataset. Most flights are labelled as on time, while a smaller share are labelled as delayed, so the data is somewhat imbalanced. This is important because a model could still achieve fairly high overall accuracy just by predicting most flights as on time. For that reason, it is also important to look at how well the model identifies delayed flights, not only its total accuracy.



Figure 1: Class balance — delayed vs on time

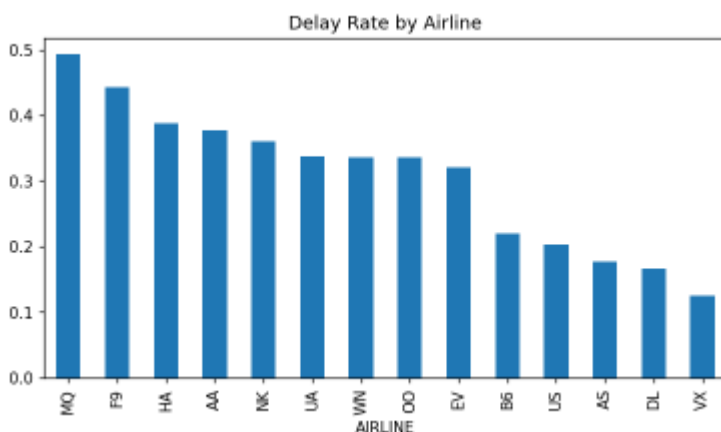


Figure 2: Delay rate (%) by airline

3. Methods

The analysis followed a standard supervised classification pipeline:

- Cancelled flights and any rows with missing values were removed to avoid issues during modelling
- Scheduled departure time (originally in HHMM format) was converted into a simpler departure hour value (0–23)
- Categorical variables such as airline, origin, and destination were converted into numbers using Label Encoder
- A binary target variable was created, where 1 represents flights delayed by more than 15 minutes and 0 represents all others
- Features were scaled using StandardScaler before applying Logistic Regression, since that model is sensitive to feature scale, while the Decision Tree was trained on the original encoded values
- The dataset was split into training and testing sets using an 80/20 split, with a fixed random state (42) to keep results consistent

Two models were compared:

- Logistic Regression: used as a basic model to check how well simple linear relationships can predict flight delays
- Decision Tree Classifier: used to capture more complex patterns, such as how different factors (like airline and time of day) combine to affect delay risk

4. Results

Figure 2 shows that delay rates are not the same across all airlines, with some carriers having noticeably higher proportions of delayed flights than others. Figure 3 shows an even clearer pattern: departure hour appears to matter a lot. The chance of delay generally increases as the day goes on, with the highest delay risk appearing later in the afternoon and evening. In contrast, early morning flights tend to be more reliable, which suggests that delays build up over the course of the day.

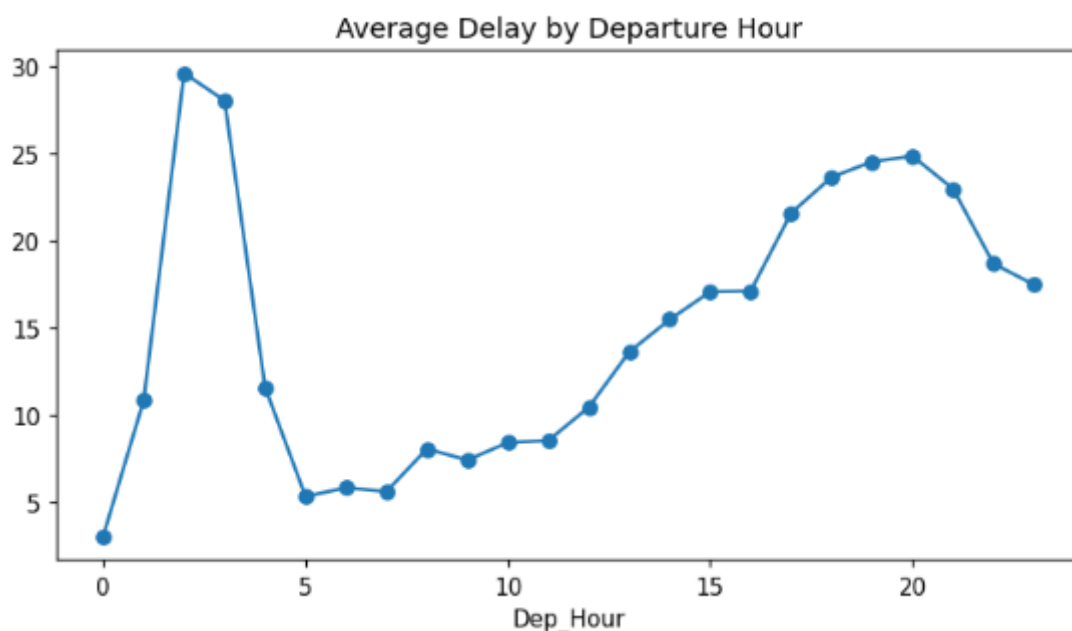


Figure 3: Delay rate (%) by scheduled departure hour — risk rises sharply from midday onwards

Model performance results:

- Logistic Regression accuracy: 0.794 — performed well overall but missed many delayed flights
- Decision Tree accuracy: 0.821 — this model gave better overall performance and also did a noticeably better job of identifying delayed flights

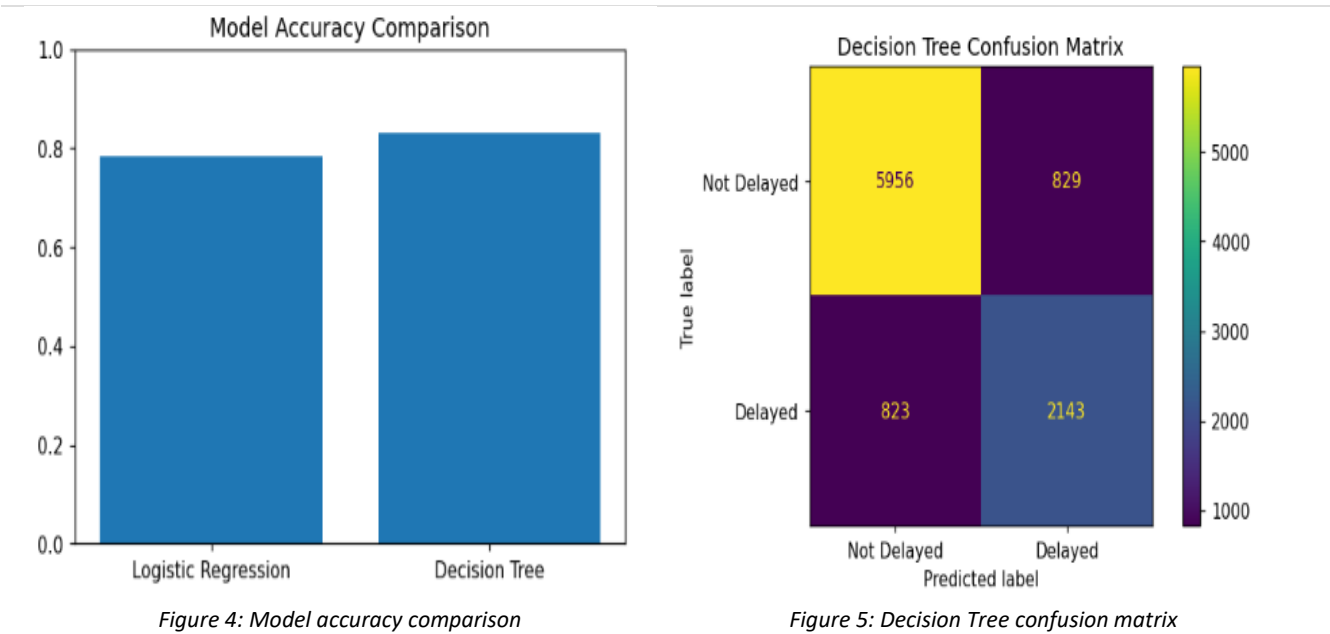
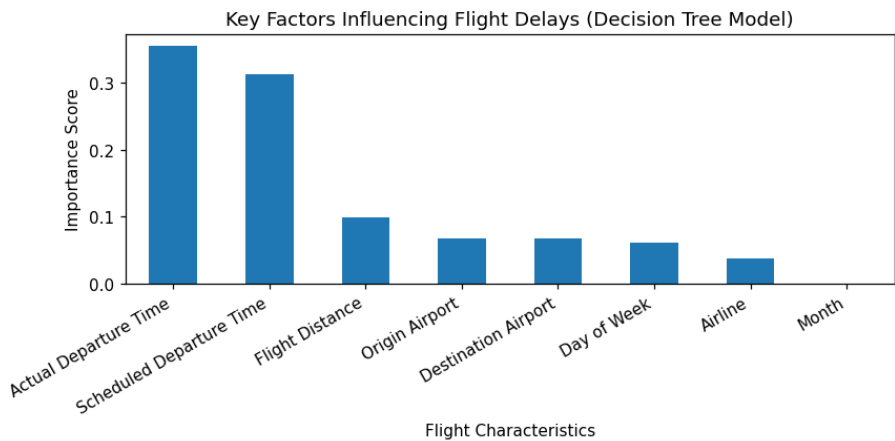


Figure 6 shows the feature importance results from the Decision Tree model. The strongest influence comes from departure-related time variables, which suggests that timing plays the biggest role in predicting delays. Distance and airport-related variables also contribute, but to a smaller extent. Airline code and calendar-based features, such as month or day of the week, appear to have less impact in this analysis.



5. Conclusions & Recommendations

This study shows that flight delays can be predicted reasonably well using pre-departure operational data. Among the two models tested, the Decision Tree performed better than Logistic Regression, suggesting that delay patterns are influenced by more complex relationships than a simple linear model can capture. The results also show that departure timing is the strongest predictor, with later flights generally facing a higher risk of delay, likely because disruptions build up over the day.

Based on these findings, airlines and airports should pay closer attention to flights scheduled in the afternoon and evening, where delays appear more likely. Extra buffer time, better crew and aircraft planning, and earlier passenger updates during these periods could help reduce knock-on disruption. This analysis could also be improved in future by adding weather and real-time airline performance data, as these are likely to affect delays as well. Overall, the results suggest that even a simple predictive model can be useful as an early warning tool, and with richer data and further model development, it could support more practical delay management in real operations.